

Practicum No15**Date: 17/4/2026****Investor Pitch Deck Outline for a CSE/AIML/DSCS/DS Start-up****I. Practical Significance**

An Investor Pitch Deck is a critical communication tool that enables technology-driven startups to present their value, innovation, and growth potential clearly and convincingly. For CSE, AIML, DSCS, and Data Science enterprises, a well-structured pitch deck helps translate complex technical ideas into investor-friendly insights. This practicum trains learners to articulate the problem, solution, and technical uniqueness of their product in a concise and compelling manner.

Students learn to highlight AI/ML capabilities, data-driven advantages, scalability, and competitive differentiation.

The practicum builds the ability to justify business models, revenue projections, and customer acquisition strategies with evidence. It also develops skills to visually communicate architecture, workflows, and performance benchmarks to non-technical stakeholders. Learners gain confidence in presenting traction metrics, market potential, and risk-mitigation strategies that influence investor decisions.

The practicum strengthens strategic thinking by linking technical features with market viability and business sustainability. By preparing a realistic and persuasive pitch deck, learners enhance their entrepreneurial readiness in the technology startup ecosystem.

Overall, this practicum equips students with the communication and analytical skills needed to attract investors and scale their AI/tech ventures.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a Pitch deck, project proposals for small/medium-scale startups logically and scientifically.
- a) **Develop** innovative and feasible startup ideas for CSE, AIML, DSCS, and Data Science domains. (PDO).
- b) **Identify** emerging opportunities in artificial intelligence, machine learning, cybersecurity, cloud computing, data analytics, and intelligent automation technologies. (CDO).
- c) **Apply** entrepreneurial traits, technical creativity, and ethical responsibility in designing investor-ready, technology-based ventures. (ADO)

III. Relevant CO(s)

- **CO5. Develop** a pitch deck of the identified product/service and its management strategies for ROI.

IV. Practical Outcome (PrO)

- **PrO15 – Present** a pitch deck outline to motivate angel investors or venture capitalists for **the identified** new enterprise.

V. Related ADO(s)

- a) **Exhibit** innovative and analytical thinking while structuring and presenting the pitch deck.
- b) **Communicate** ideas effectively within teams and during investor-focused presentations.
- c) **Demonstrate** ethical decision-making while proposing technology-driven startup solutions..

VI. Minimum Underpinning Theory

An investor pitch deck is a strategic communication tool used to present the value, potential, and viability of a startup to prospective investors. In the domains of Computer Science (CSE), Artificial Intelligence & Machine Learning (AIML), Data Science & Cybersecurity (DSCS), and Data Science (DS), pitch decks must effectively translate complex technical innovation into clear business value. Investors typically evaluate feasibility based on technological uniqueness, scalability, market relevance, and revenue potential.

Key theoretical concepts include:

- **Problem–Solution Fit:** Identifying a meaningful technical or business problem and demonstrating how the AI/ML or data-driven solution addresses it more effectively than existing alternatives.
- **Value Proposition Design:** Clearly articulating the product’s promised value using frameworks such as the Value Proposition Canvas, highlighting efficiency, accuracy, automation, or security advantages.
- **Business Model Theory:** Applying models such as SaaS, subscription, usage-based pricing, or enterprise licensing to explain revenue generation and long-term sustainability.
- **Market Analysis & TAM–SAM–SOM Framework:** Assessing the target market size, customer segments, and realistic adoption potential using structured market-mapping techniques.
- **Competitive Positioning:** Using tools such as SWOT, Porter’s Five Forces, and competitor grids to justify differentiation based on technology, performance metrics, or user experience.
- **Financial Planning:** Incorporating fundamental finance principles such as cost forecasting, breakeven analysis, unit economics, and ROI expectations to strengthen investor confidence.
- **Technology Readiness & Scalability:** Demonstrating product maturity using TRL (Technology Readiness Levels), scalability models, and cloud/AI deployment frameworks.
- **Go-to-Market (GTM) Strategy:** Applying marketing and distribution theories to show how the product will reach and retain customers through channels like SaaS platforms, developer communities, and enterprise sales.

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and documentation	1 per team
2	Internet access	For identifying trends and problems	As required
3	Brainstorming worksheet / chart paper	For idea listing and mind mapping	1 set
4	Reference materials	Magazines, journals, websites	1 per team

IX. Safety Precautions

- a) Use only original content or properly referenced material to prevent plagiarism.
- b) Avoid using confidential or proprietary data.
- c) Follow ethical and social boundaries while building a startup.

X. Procedure

1. Identify the startup idea and define the problem it aims to solve.
2. Identify the market, customer segment, and competition thoroughly.
3. Create the value proposition, product features, and technology overview.
4. Prepare business model elements such as revenue streams and cost structure.
5. Develop market sizing and growth opportunity visuals.
6. Include financial forecasts and investment requirements.
7. Design clear, concise, and visually appealing pitch deck slides.

XI. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	ASUS	Intel i5/i7 processor, 8GB RAM, Windows 11 used for preparing the pitch deck	1	-
2.	Canva (Web Tool)	Free	Online presentation tool used for designing pitch deck templates and slides	1	-
3.	Internet Connection	Wi-Fi	Used for research, accessing Canva, and gathering market information	As required	-
4.	Reference Materials	Online	Canva templates, startup pitch deck examples, YouTube tutorials, and textbooks	As required	-

XII. Actual Procedure Followed

1. Selected the startup idea **Digital Attendance and Productivity Tracker** based on the mini-project concept.
2. Studied the purpose and structure of an **investor pitch deck**, which is used to present startup ideas to potential investors.
3. Chose a suitable **pitch deck template using Canva** to design the presentation slides.
4. Organized the slides to include important sections such as **problem statement, proposed solution, product and technology overview, target market, and value proposition**.
5. Added business-related information including **business model, market size estimation, competitor analysis, marketing strategy, and financial highlights**.
6. Designed the slides using **icons, charts, and consistent colour themes** to improve visual clarity and presentation quality.
7. Ensured that each slide contained **short bullet points and clear explanations** so the startup idea could be easily understood.
8. Reviewed and refined the pitch deck to improve **content clarity, logical flow, and overall presentation quality**.

XIII. Observations and Data

S. No.	Entrepreneur	Major Venture	Key Traits	Opportunity Identified
1	Elon Musk	Tesla, SpaceX	Visionary thinking, risk-taking	Electric mobility and space technology
2	N. R. Narayana Murthy	Infosys	Integrity, innovation	Global IT outsourcing and software services
3	Ritesh Agarwal	OYO Rooms	Persistence, creativity	Budget hospitality and smart booking platforms
4	Byju Raveendran	BYJU'S	Tech-savvy, empathetic	Online education and EdTech solutions
5	Falguni Nayar	Nykaa	Market insight, resilience	Beauty e-commerce and retail technology

XIV. Results/Observations

1. A 13-slide investor pitch deck was prepared for the Digital Attendance and Productivity Tracker startup.
2. The presentation included key sections such as problem statement, proposed solution, product overview, target users, value proposition, business model, market analysis, competitor comparison, marketing strategy, financial highlights, and funding requirements.
3. The pitch deck was designed using Canva templates with a clean and professional layout.
4. Visual elements such as icons, diagrams, and charts were used to improve readability and presentation clarity.
5. The pitch deck clearly communicated the technical concept, market need, and potential business value of the startup idea.

XV. Interpretation of Results

The prepared pitch deck demonstrates that a **clear and structured presentation** is important for explaining a startup idea to investors. By presenting the problem, solution, and business model in a concise way, the startup concept becomes easier to understand and evaluate.

Using frameworks such as **market analysis, competitor comparison, and financial projections** helps increase the credibility of the startup proposal. A strong value proposition and clear target market improve the chances of attracting investor interest.

XVI. Conclusions

The **Digital Attendance and Productivity Tracker** startup concept has strong potential for practical implementation in educational institutions. Preparing the investor pitch deck helped clearly define the **problem, solution, market opportunity, and business model**.

This practicum demonstrated that a **well-designed and visually appealing pitch deck** is an essential tool for communicating startup ideas to investors and stakeholders. With further improvements such as real market validation and detailed financial projections, the startup idea can be prepared for potential funding opportunities.

XVII. Suggested Practicum Related Questions *(Every batch will be given another set of questions)*

1. **Evaluate** the clarity and impact of the problem statement presented in the pitch deck.
2. **Analyse** the strength of the value proposition in differentiating the product from competitors.
3. **Assess** whether the market sizing and customer segmentation are realistic and data-driven.
4. **Examine** the feasibility of the financial projections and revenue model.
5. **Justify** the investment amount requested based on technology readiness and scalability.
6. **Compare** the competitive analysis and the startup's positioning strategy.
7. **Recommend** improvements in the pitch deck's visual communication and storytelling flow.
8. **Predict** investor concerns related to technology risks or market adoption.
9. **Propose** strategic partnerships that can strengthen the startup's growth potential.
10. **Formulate** enhancements to the go-to-market plan to improve customer acquisition.

XVIII. References / Suggestions for further reading

Timmons & Spinelli, New Venture Creation: Entrepreneurship for the 21st Century, McGraw-Hill, 2019

XIX. Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the **process-related skills** and **product-related skills** will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Identification of focus domain and problem areas	3	
2	Use of structured idea generation techniques	4	
3	Team collaboration and creativity	4	
4	Feasibility evaluation and reasoning	3	
5	Ethical originality and clarity of ideas	4	
Product-related Skills: 12 Marks - 40%			
1	Quality and novelty of generated ideas	3	
2	Logical presentation and organization	3	
3	Clarity of findings and conclusions	3	
4	Report/presentation submission	3	
5	Innovation potential and market relevance	3	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Signature with Date
Process-Related skills (18)	Product-Related skills (12)	Total (30)		